

Optimal Transcutaneous Electrical Stimulation (tES) Cure Protocols for Gangrene: A Finite Mathematical Framework using QML and GenAI

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Abstract: Treating complex peripheral tissue degradation, such as nicotine-induced gangrene, demands adaptive interventional therapeutics. This report establishes a finite mathematical framework detailing transcutaneous electrical stimulation (tES) protocols. By combining discrete probabilistic mapping via Quantum Machine Learning (QML) and Hebbian modifications driven by Generative AI, we define precise tissue viability and optimal waveform topologies.

1. Finite Quantum Machine Learning Probabilistic Maps

To compute comparative differential estimates for removing dead tissue while retaining healthy tissue, a discrete probabilistic mapping is applied over the finite grid i,j . The tissue viability state V is calculated using a recursive energy expectation function:

$$P(V_{i,j} = \text{healthy}) = 1 / [1 + \exp(-\gamma \cdot \Delta E_{i,j})]$$

where $\Delta E_{i,j}$ represents the local necrotic risk measure factored against quantum Hamiltonian estimators. We use a finite difference Laplacian to identify sharp boundaries between viable and gangrenous clusters, minimizing unpredicted surgical excision.

2. Generative AI Finite Hebbian Waveform Modifications

To induce targeted necrotic reversal, electrical stimulation pulses are iteratively refined. The Generative AI model updates the temporal stimulation envelope $W(t)$ over N discrete steps utilizing finite Hebbian plasticity rules:

$$W_{k+1} = W_k + \alpha \cdot \Delta t \cdot [S_k(t) \cdot R_k(t) - \lambda W_k]$$

where S represents the applied pulse current, R is the measured electrical impedance response of the recovering tissue, and α governs the GenAI learning rate. The decay parameter λ guarantees the waveforms remain within safe physiological limits to prevent thermal burns.

3. Clinical Optimization and Diffeomorphic Energy Constraints

Total therapeutic energy deposited over the discrete protocol time bounds $T = N\Delta t$ is dynamically controlled. The energy integration follows a strict Euler finite bounding formulation:

$$E_{total} \approx \sum (W_k^2 / Z_k) \cdot \Delta t \leq E_{threshold}$$

By maintaining E_{total} strictly below the cell wall degradation limits while optimizing temporal pulse distributions via QML metrics, these protocols significantly out-perform static waveforms in promoting vascular recovery.